

LP modeling

• Basic steps in LP model formulation

1 Determine the decision variables

- A decision variable is both under the control of the decision maker and could have an impact on the solution to the problem of interest.

2 Formulate the objective function

- An objective is represented by a function of the decision variables. Such a function usually represents the desires of the decision maker.

eg maximize profit

min costs

max resource utilization

min machine downtime

min risk

3 Formulate the constraints

- A constraint is an equation or inequality that represents a restriction due to a resource or technological limitation.

eg limited raw material

budget

time

personnel

ability / skills

1 A Brewery Problem

Microbrewers Incorporated makes four products called Light, Dark, Ale, and Premium. These products are made using the resources of water, malt, hops, and yeast. Microbrewers has a free supply of water, so it is the amount of the other resources that restricts production capacity. The technology table gives the pounds of each resource required in the production of 1 gal of each product. The problem faced by Microbrewers Inc. is to decide how much of each product it should make in order to maximize its revenue.

	Light	Dark	Ale	Premium	Available
Malt	1	1	0	3	50 lb
Hops	2	1	2	1	150 lb
Yeast	1	1	1	4	80 lb
Revenue	\$6	\$5	\$3	\$7	

Table 1: Technology Table for Microbrewers, Inc.

1. x_1 = amount of Light produced

x_2 = Dark ...

x_3 = Ale ...

x_4 = Premium ...

2. max $6x_1 + 5x_2 + 3x_3 + 7x_4$

3. s.t. $x_1 + x_2 + 3x_4 \leq 50$ (Malt)

$2x_1 + x_2 + 2x_3 + x_4 \leq 150$ (Hops)

$x_1 + x_2 + x_3 + 4x_4 \leq 80$ (Yeast)

$x_1, x_2, x_3, x_4 \geq 0$ (nonnegativity)

2 A Product-Mix Problem

HiTech, Inc., a small manufacturing firm produces two microwave switches, Switch A and Switch B. The return per unit of Switch A is \$20, whereas the return per unit of Switch B is \$30. Because of contractual commitments, HiTech must manufacture at least 25 units of Switch A per week, and based on the present demand for its products, it can sell all that it can manufacture. However, it wishes to maximize profit while determining the production sizes to satisfy various limits resulting from a small production crew. These include

- ③ { Assembly hours: 240 hours available per week
Testing hours: 140 hours available per week

Switch A requires 4 hours of assembly and 1 hour of testing, and Switch B requires 3 hours and 2 hours, respectively.

1 x_1 = amount of switch A produced

x_2 = - - - B -

2	max	$Z = 20x_1 + 30x_2$	(profit per week)
3	s.t.	$x_1 \geq 25$	(contractual commitments)
		$4x_1 + 3x_2 \leq 240$	(assembly hours)
		$x_1 + 2x_2 \leq 140$	(testing hours)
		$x_1, x_2 \geq 0$	(nonnegativity)

Note: In fact, this problem should be modeled as an integer linear program.

That is, x_1, x_2 are supposed to be integers.

3 An Investment-Planning Problem

An investor has ^① decided to invest a total of \$50,000 among three investment ^③ opportunities: savings certificates, municipal bonds, and stocks. The annual ^② return on each investment is estimated to be 7%, 9%, and 14%, respectively. The investor does not intend to invest his annual interest returns (that is, he plans to use the interest to finance his desire to travel). ^④ He would like to maximize his yearly return while ^⑤ investing a minimum of \$10,000 in bonds. Also, the investment in stocks should not exceed the combined total investment in bonds and savings certificates. And, finally, he should invest between \$5,000 and \$15,000 in savings certificates.

$\rightarrow x_2 \geq 10,000$
 $\rightarrow x_3 \leq x_1 + x_2$
 $\rightarrow 5,000 \leq x_1 \leq 15,000$

- 1 $x_1 = \$$ invested in savings certificates
- $x_2 =$ ' municipal bonds
- $x_3 =$ ' stocks

- 2 $\max Z = 0.07x_1 + 0.09x_2 + 0.14x_3$

- 3 s.t. $x_1 + x_2 + x_3 = 50,000$

$$x_2 \geq 10,000$$

$$-x_1 - x_2 + x_3 \leq 0$$

$$x_1 \geq 5,000$$

$$x_1 \leq 15,000$$

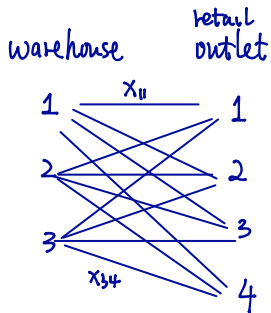
$$x_1, x_2, x_3 \geq 0$$

4 A Transportation Problem

A manufacturer has three warehouses that supply finished product to four retail outlets. ^③The warehouses have 6000, 9000, and 4000 units available, and the demands at the retail outlets are projected to be 3900, 5200, 2700, and 6400 units. The per unit costs (in dollars) of shipping from each warehouse to each retail outlet are given in Table 2. ^②The manufacturer needs to determine the minimum-cost shipping schedule that satisfies all demands. ^①

	retail outlets			
warehouse	1	2	3	4
1	7	3	8	4
2	9	5	6	3
3	4	6	9	6

Table 2: Unit Shipping Costs

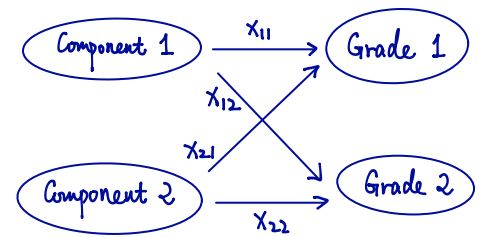


1. x_{ij} = quantity shipped from Warehouse i to Retail Outlet j $i=1,2,3$, $j=1,2,3,4$

(12 variables)

$$\begin{aligned}
 2 \quad \min \quad & Z = 7x_{11} + 3x_{12} + 8x_{13} + 4x_{14} + 9x_{21} + 5x_{22} + 6x_{23} + 3x_{24} + 4x_{31} + 6x_{32} + 9x_{33} + 6x_{34} \\
 3 \quad \text{s.t.} \quad & x_{11} + x_{12} + x_{13} + x_{14} = 3900 \quad (\text{Retail Outlet 1}) \quad [\text{Demand}] \\
 & x_{12} + x_{22} + x_{32} = 5200 \quad (2) \\
 & x_{13} + x_{23} + x_{33} = 2700 \quad (3) \\
 & x_{14} + x_{24} + x_{34} = 6400 \quad (4) \\
 & x_{11} + x_{12} + x_{13} + x_{14} \leq 6000 \quad (\text{Warehouse 1}) \quad [\text{Supply}] \\
 & x_{21} + x_{22} + x_{23} + x_{24} \leq 9000 \quad (2) \\
 & x_{31} + x_{32} + x_{33} + x_{34} \leq 4000 \quad (3) \\
 & x_{ij} \geq 0 \quad (i=1,2,3, j=1,2,3,4)
 \end{aligned}$$

1 x_{11} = # barrels of Component 1 blended into Grade 1
 x_{12} = 1 -- 2
 x_{21} = 2 1
 x_{22} = 2 - 2



5 A Product-Blending Problem

- ② The Sierra Refining Company produces two grades of unleaded gasoline, Grade 1 and Grade 2, which it supplies to its chain of service station for \$48 and \$53 per barrel, respectively. Both grades of gasoline are blended from Sierra's inventory of gasoline components and must meet the specifications in Table 3. The characteristics of the components in inventory are found in Table 4.

gasoline	min octane rating	max demand (barrels/wk)	min. deliveries (barrels/wk)
Grade 1	87	80,000	60,000
Grade 2	93	40,000	15,000

Table 3: Gasoline Specifications

gasoline component	octane rating	inventory (barrels)	② cost (\$/barrel)
1	86	70,000	33
2	96	60,000	37

Table 4: Components Characteristics

- ① What quantities of the two components should be blended into the two gasolines in order to maximize weekly profit?

Hint: We need to assume that octanes blend linearly. That is, the octane level in a mixture of components is obtained by computing the weighted average of the octane in the mixture.

2 Products Grade 1 $x_{11} + x_{21}$
 Grade 2 $x_{12} + x_{22}$

Revenue $48(x_{11} + x_{21}) + 53(x_{12} + x_{22})$

Components 1 $x_{11} + x_{12}$
 2 $x_{21} + x_{22}$

Cost $33(x_{11} + x_{12}) + 37(x_{21} + x_{22})$

profit = revenue - cost

max $z = 15x_{11} + 20x_{12} + 11x_{21} + 16x_{22}$

3 min octane rating

Grade 1 $\frac{86x_{11} + 96x_{21}}{x_{11} + x_{21}} \geq 87 \rightarrow$

Grade 2 $\frac{86x_{12} + 96x_{22}}{x_{12} + x_{22}} \geq 93 \rightarrow$

max demands Grade 1
 Grade 2

min deliveries Grade 1

2
5

inventories (supply constraints)

Component 1

Component 2

nonnegativity

s.t.

$-x_{11} + 9x_{21} \geq 0$

$-7x_{12} + 3x_{22} \geq 0$

$x_{11} + x_{21} \leq 80,000$

$x_{12} + x_{22} \leq 40,000$

$x_{11} + x_{21} \geq 60,000$

$x_{12} + x_{22} \geq 15,000$

$x_{11} + x_{12} \leq 70,000$

$x_{21} + x_{22} \leq 60,000$

$x_{11}, x_{12}, x_{21}, x_{22} \geq 0$